

Acoustic Stack Optimization and Clinical Validation Preparation of a Flexible PMN-PT Ultrasonic Transducer Array for Wearable Blood Pressure Monitoring

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Background, Motivation and Objective

Wearable ultrasonic transducer arrays (UTA) enable continuous, noninvasive vascular monitoring; however, acoustic inefficiencies caused by impedance mismatch and backward wave reflections impacts bandwidth, beam symmetry, and waveform fidelity. Shown in Figure 1(a) is our previously developed flexible 5 × 4 PMN-PT 1–3 composite array that operates at 6 MHz and demonstrates accurate vessel tracking, yet performance remains limited without acoustic stack optimization. The objective of this work is twofold: (1) to enhance acoustic transmission and reception characteristics through matching and backing layers, and (2) to prepare the optimized system for medical validation using a programmable ultrasound research platform.

Statement of Contribution/Methods

A flexible 5 × 4 PMN-PT 1–3 composite UTA fabricated using low-temperature Sn–Bi eutectic bonding was integrated with tailored matching (ML) and backing layers (BL). The BL (0.5 mm) was characterized to evaluate acoustic delay and attenuation, while ML samples (0.1–0.4 mm) were synthesized to improve impedance transition between the piezocomposite and soft tissue. Acoustic characterization, speed, and attenuation were measured using an A309S transducer, HGL hydrophone, oscilloscope, and 3-axis scanning platform as depicted in Figure 1(b). Beam transmission, directivity, and receiving sensitivity were evaluated experimentally. For validation preparation, the optimized UTA was interfaced with the Verasonics Vantage system for RF-based vessel tracking in a vascular phantom, the setup is shown in Figure 1(c). Diameter extraction was performed via time-of-flight analysis and pressure was reconstructed using an exponential pressure–area model.

Results/Discussion

The incorporation of ML and BL improved forward acoustic transmission, reduced ringing, and increased bandwidth by approximately 1.7 MHz. Beam measurements showed a narrower and more symmetric main lobe with reduced side-lobe energy compared to the no-layer configuration, indicating behavior closer to an ideal piston source. Receiving sensitivity remained stable across angular variation. Using the Vantage 64 - Verasonics platform, phantom-based blood pressure reconstruction yielded systolic and diastolic errors of approximately ~4 mmHg and ~2 mmHg relative to a reference sensor. These findings demonstrate that acoustic stack optimization can significantly enhance signal fidelity while establishing readiness for translational and pilot on-body evaluation of wearable blood pressure monitoring systems.

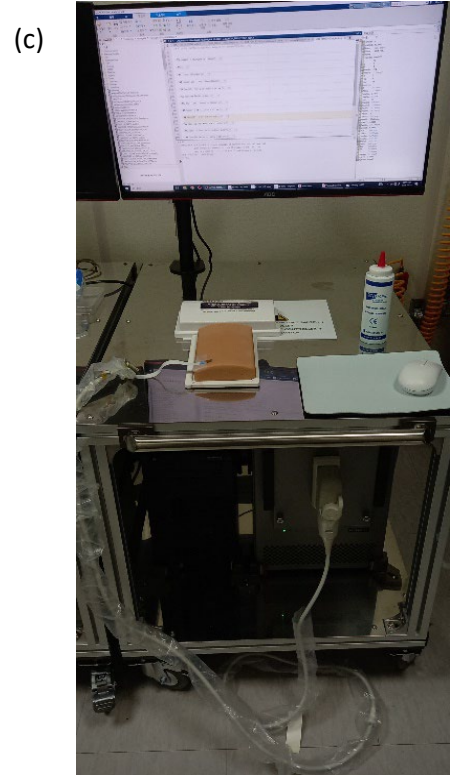
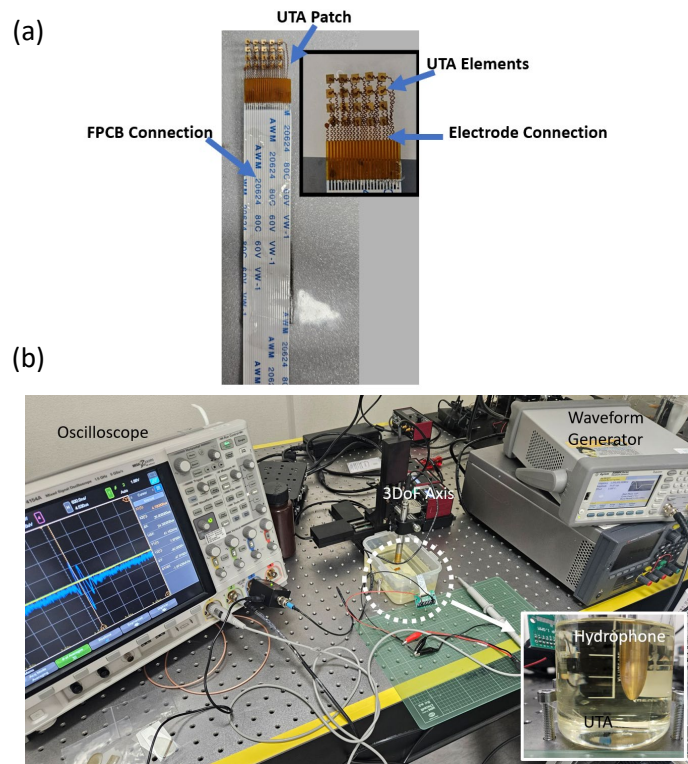


Figure 1. (a) Fabrication of the flexible and stretchable UTA showing only the elements and connection, (b) Acoustic characterization of the UTA with matching and backing layer, (c) Setup of the Verasonics system with UTA and target phantom